Free multi-Novikov algebras as canonical multidifferential subalgebras (Theorem 3.3)

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Editorial note. Native statement and proof are unchanged. Selected bibliography reconstructed only from the same official published PDF.

Editorial scope. Author-material preparation only; one Theorem 3.3, no mathematical repair or formal admission.

Allowed standard prerequisites (editorial).

- Knuth's rotation correspondence. [Knu97], The Art of Computer Programming, Vol. 1, Fundamental Algorithms, 3rd ed. (1997), Chapter 2.3. Standard planar binary-tree/rooted-tree correspondence; actual map, labeling rule and both native diagrams retained at lines 430–492.
- Free commutative differential algebra and graded dimensions. Definition 3.1, Proposition 3.2, and the finite-set/finer-grading footnote of this source; [Bur06] Section 2.1 provides comparative context. Polynomial realization is stated in full, commuting derivations checked in Proposition 2.1; finite-dimensional surjection/dimension comparison is elementary linear algebra.
- Earlier single-operation Novikov realization. [DL02] Theorem 7.8; [BN85], [Osb92]; [BCCH21] Proposition 4.21. Earlier realization is acknowledged strategy/context, not a substitute for current proof. All new multi-operation normal-form, spanning and bijection arguments are retained.
- General multi-index setting. [BL24], published bibliography key BL23. Background motivation for Section 2 variables and operations; target Theorem 3.3 has its own complete source-local algebraic proof.

Source and counted unit. Yvain Bruned and Vladimir Dotsenko, *Novikov algebras and multi-indices in regularity structures*, Annales Henri Lebesgue 9 (2026), 791–814, `https://doi.org/10.5802/ahl.275`. Published under CC BY 4.0: `https://creativecommons.org/licenses/by/4.0/`.
The counted unit is the whole Theorem 3.3 (free multi-Novikov realization). The canonical-structure proof, populated-monomial and dimension bounds, ordered and normal reductions, and both native rotation-correspondence diagrams are support for that one unit. Sections 2 and 3 below preserve the original numbering; no helper proposition is separately counted.
Editorial boundary. The original author mathematics is retained, including its notation and wording. The journal class and branding are replaced by a standard wrapper. The seven cited bibliography entries are transcribed from the published bibliography, pp.812–813, with original printed labels; no missing BibTeX file is guessed. Knuth's rotation correspondence and elementary free differential-algebra/graded dimension facts are standard prerequisites; the current multi-operation reductions remain fully reproduced.
The source's finite-set working assumption and its infinite-set grading footnote are retained. In particular the source uses "right-leaning" within the proof of Proposition 3.8 after defining "left-leaning", and its Proposition 3.4 proof uses the symbols (f_0, x_0) and (x, m_1). These are retained as printed, not silently repaired or independently certified.

2 General multi-indices

In applications of the algebraic formalism developed in this paper, one deals with stochastic PDEs of the form

$$(\partial_t - \mathcal{L})\, u = \sum_{\mathfrak{l} \in \mathfrak{L}^-} a^{\mathfrak{l}}(\mathbf{u}) \xi_{\mathfrak{l}}, \tag{2.1}$$

where u is a function of $d+1$ variables $t = x_0$, x_1, ..., x_d, $\mathcal{L}$ is a differential operator in the variables x_1, ..., x_d, $\mathfrak{L}^-$ is a finite set and the $\xi_{\mathfrak{l}}$ are space-time noises (random distributions). The bold letter $\mathbf{u}$ in each $a^{\mathfrak{l}}(\mathbf{u})$ means that these are functions of u and of its iterated partial derivatives. For $\mathbf{n} = (n_0, \dots, n_d) \in \mathbb{N}^{d+1}$, one may consider the expressions

$$u^{(\mathbf{n})} := \frac{\partial_{x_0}^{n_0} \dots \partial_{x_d}^{n_d}}{n_0! \dots n_d!}(u),$$

which are the coefficients of the Taylor expansion of u around the point x. To each $a^{\mathfrak{l}}(\mathbf{u})$ one may apply the derivatives $\partial_{u^{(\mathbf{n})}}$ which pairwise commute. In this context, one typically works with polynomials in the expressions

$$\left(\prod_{\mathbf{n} \in \mathbb{N}^{d+1}} \partial_{u^{(\mathbf{n})}}^{w_{\mathfrak{l}}(\mathbf{n})} \right) (a^{\mathfrak{l}}(\mathbf{u})),$$

where $w_{\mathfrak{l}}(\mathbf{n}) \in \mathbb{N}$ are the orders of derivatives applied to $a^{\mathfrak{l}}(\mathbf{u})$. These expressions will be now encoded in a formal algebraic way via what we shall call the general multi-indices. For a set I, we define the set of multi-indices over I, denoted by $M(I)$, as the set of all finitely supported maps $w \colon I \to \mathbb{N}$, that is the maps for which all but finitely many $i \in I$ are mapped to 0. We shall now consider $M(\mathbb{N}^{d+1})$, multi-indices over $\mathbb{N}^{d+1}$, which one can regard as commutative monomials in letters $\mathbf{n} \in \mathbb{N}^{d+1}$.

Let us now introduce formal variables $z_{(\mathfrak{l},w)}$, $(\mathfrak{l}, w) \in \mathfrak{L}^- \times M(\mathbb{N}^{d+1})$, and define the *general multi-indices* β as monomials

$$z^{\beta} := \prod_{(\mathfrak{l},w) \in \mathfrak{L}^- \times M(\mathbb{N}^{d+1})} z_{(\mathfrak{l},w)}^{\beta(\mathfrak{l},w)}. \tag{2.2}$$

These multi-indices have been introduced in the recent work [BL24]. The vector space spanned by all monomials representing general multi-indices is just the polynomial algebra in variables $z_{(\mathfrak{l},w)}$, $(\mathfrak{l}, w) \in \mathfrak{L}^- \times M(\mathbb{N}^{d+1})$. That algebra has, for each $\mathbf{n} \in \mathbb{N}^{d+1}$, the derivation $D^{(\mathbf{n})}$ defined by the formula

$$D^{(\mathbf{n})} = \sum_{(\mathfrak{l},w) \in \mathfrak{L}^- \times M(\mathbb{N}^{d+1})} (w(\mathbf{n}) + 1)\, z_{(\mathfrak{l},w+e_{\mathbf{n}})} \partial_{z_{(\mathfrak{l},w)}}. \tag{2.3}$$

On the original level of polynomials in iterated partial derivatives of $a^{\mathfrak{l}}(\mathbf{u})$, this derivation corresponds to applying one extra derivative $\partial_{u^{(\mathbf{n})}}$; as in the formula for the derivation D in the introduction, the coefficient $w(\mathbf{n}) + 1$ here corresponds to one of the possible choices of a basis.

We shall be only interested in meaningful multi-indices called *populated* satisfying the condition

$$\sum_{(\mathfrak{l},w)} (1 - |w|))\beta(\mathfrak{l}, w) = 1 \tag{2.4}$$

where $|w|$ is the degree of w, viewed as a commutative monomial in letters $\mathbf{n} \in \mathbb{N}^{d+1}$. The vector space spanned by all populated indices is denoted $\mathcal{M}^{\mathrm{gen}}$. We can define products $\triangleright_{\mathbf{n}}$, $\mathbf{n} \in \mathbb{N}^{d+1}$, on $\mathcal{M}^{\mathrm{gen}}$ by setting

$$z^{\beta} \triangleright_{\mathbf{n}} z^{\beta'} = z^{\beta} D^{(\mathbf{n})}(z^{\beta'}). \tag{2.5}$$

Our goal now is to have a better understanding of this family of products.

Proposition 2.1. *Let A be a set and R be a commutative associative algebra, and suppose that ∂_a, $a \in A$, are pairwise commuting derivations of that algebra. The bilinear operations $\triangleright_a$ on R defined by the formulas*

$$x \triangleright_a y := x \partial_a(y)$$

satisfy the identities

$$\begin{aligned}
(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) &= (y \triangleright_a x) \triangleright_b z - y \triangleright_a (x \triangleright_b z), \\
(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) &= (x \triangleright_b y) \triangleright_a z - x \triangleright_b (y \triangleright_a z), \\
(x \triangleright_a y) \triangleright_b z &= (x \triangleright_b z) \triangleright_a y,
\end{aligned}$$

for all $a, b \in A$.

Proof. Note that

$$(x \triangleright_a y) \triangleright_b z = x\partial_a(y)\partial_b(z) = x\partial_b(z)\partial_a(y) = (x \triangleright_b z) \triangleright_a y,$$

so the last identity holds. Moreover,

$$x \triangleright_a (y \triangleright_b z) = x\partial_a(y\partial_b(z)) = x\partial_a(y)\partial_b(z) + xy\partial_a\partial_b(z),$$

so

$$(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) = -xy\partial_a\partial_b(z),$$

which is symmetric in x, y (since R is commutative) and in a, b (since the derivations ∂_a commute between themselves), proving the first two identities. □

If the set A consists of one element a, these identities reduce to the identities

$$(x \triangleright_a y) \triangleright_a z - x \triangleright_a (y \triangleright_a z) = (y \triangleright_a x) \triangleright_a z - y \triangleright_a (x \triangleright_a z),$$
$$(x \triangleright_a y) \triangleright_a z = (x \triangleright_a z) \triangleright_a y,$$

which are the identities of the so-called (left) Novikov algebras [BN85, DL02, Osb92]. It is thus natural to give the following definition.

Definition 2.2. Let A be a set. A *(left) A-multi-Novikov algebra* is a vector space equipped with bilinear products $x, y \mapsto x \triangleright_a y$, $a \in A$, satisfying the identities

$$(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) = (y \triangleright_a x) \triangleright_b z - y \triangleright_a (x \triangleright_b z), \tag{2.6}$$
$$(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) = (x \triangleright_b y) \triangleright_a z - x \triangleright_b (y \triangleright_a z), \tag{2.7}$$
$$(x \triangleright_a y) \triangleright_b z = (x \triangleright_b z) \triangleright_a y, \tag{2.8}$$

for all $a, b \in A$.

Note that the first two of the multi-Novikov identities imply the identity

$$(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) = (y \triangleright_b x) \triangleright_a z - y \triangleright_b (x \triangleright_a z),$$

which is the defining identity of left multi-pre-Lie algebras [BCCH21, Proposition 4.21].

3 Free multi-Novikov algebras

The only source of multi-Novikov algebras that we have seen so far is given by Proposition 2.1, which suggests to consider the following algebraic structure.

Definition 3.1. Let A be a set. A commutative multidifferential algebra is a vector space R equipped with a commutative associative product and endomorphisms ∂_a, $a \in A$, which are pairwise commuting derivations of the product on R.

Note that it is more common to use this structure to define a single binary operation on the different vector space $R \otimes \mathrm{Vect}(A)$, see, for instance [Bur06, Section 2.1]; in our work, we use it to define many operations on the same vector space R.

Using this terminology, we may re-state Proposition 2.1 by saying that every commutative multidifferential algebra has a natural multi-Novikov structure, which we shall call the *canonical multi-Novikov structure*.

The following almost obvious proposition describes the free commutative multidifferential algebra generated by a set X, which we shall denote $\mathsf{CD}_A\langle X\rangle$. For convenience of the reader, we recall that the notation $M(-)$ refers to multi-indices, that is finitely supported maps to $\mathbb{N}$.

Proposition 3.2. *The algebra $\mathsf{CD}_A\langle X\rangle$ is isomorphic to the polynomial algebra on the set of generators $X \times M(A)$ equipped with the derivations ∂_a defined on generators as follows: for $(x, m) \in X \times M(A)$, we have*

$$\partial_a(x, m) = (x, m_a),$$

where $m_a(a') = m(a') + \delta_{a,a'}$. In other words, the generator (x, m) of the polynomial algebra is identified with the element $\prod_{a\in A} \partial_a^{m(a)}(x)$ of $\mathsf{CD}_A\langle X\rangle$.

The main result of this section is the following theorem that describes the free multi-Novikov algebra generated by a set X, which we shall denote $\mathsf{Nov}_A\langle X\rangle$. This theorem is a generalisation of [DL02, Theorem 7.8], and follows a similar strategy; however, given how intricate the proof is, we deemed important to give it with all the details.

Theorem 3.3. *The algebra $\mathsf{Nov}_A\langle X\rangle$ is isomorphic to the multi-Novikov subalgebra of the free commutative multidifferential algebra $\mathsf{CD}_A\langle X\rangle$ generated by X; here we consider $\mathsf{CD}_A\langle X\rangle$ with its canonical multi-Novikov structure.*

The proof of this theorem consists of several parts. First, let us describe the multi-Novikov subalgebra of the free commutative multidifferential algebra $\mathsf{CD}_A\langle X\rangle$ generated by X in a more concrete way.

Proposition 3.4. *The multi-Novikov subalgebra of the free commutative multidifferential algebra $\mathsf{CD}_A\langle X\rangle$ generated by X is spanned by all monomials $\prod_{x,m}(x,m)^{n_{x,m}}$ for which the "populated condition" holds:*

$$\sum_{x,m} n_{x,m}\left(-1+\sum_{a\in A} m(a)\right) = -1.$$

Proof. We shall freely use the identification of generators $x \in X$ with $(x,0)$ in $\mathsf{CD}_A\langle X\rangle$. Note that the populated condition holds for each generator $x \in X$: we have $m(a) = 0$ for all $a \in A$ and $n_{x,m} = 1$. Moreover, if we have two monomials u, v for which the populated condition holds, then $u \triangleright_a v = u\partial_a(v)$ is a sum of monomials for which the populated condition holds as well, so the vector space spanned by such monomials is a multi-Novikov subalgebra. Let us show that this subalgebra is generated by X. Let us take some monomial u satisfying the populated condition. If for some $a \in A$ and some $x \in X$ we have $u = \partial_a(x)u_1$, we have $u = u_1 \triangleright_a x$, and we may argue by induction on degree. Otherwise, u is a product of a monomial in variables X and a monomial in variables (x,m) where $\sum_{a\in A} m(a) \geq 2$. Let us take one of the latter, say (x_0,m_0) with $\sum_{a\in A} m_0(a) = d \geq 2$, and take some a_0 for which $m_0(a_0) \neq 0$. Because of the populated condition, the monomial u is also divisible by a monomial of degree at least d in variables from X, say $u = x_{i_1}\dots x_{i_d}(f_0,x_0)u_1$. Note that the populated condition is satisfied for u_1. Let us define $m_1 \in M(A)$ by the formula $m_1(a) = m_0(a) - \delta_{a_0,a}$. Note that the populated condition is satisfied for $x_{i_1}\dots x_{i_d}(x,m_1)$, and we have

$$u_1 \triangleright_{a_0} (x_{i_1}\dots x_{i_d}(x,m_1)) = \partial_{a_0}(x_{i_1}\dots x_{i_d}(x,m_1))u_1 = \\ u + \sum_{j=1}^{d}(x_{i_1}\dots x_{i_{j-1}}x_{i_{j+1}}\dots x_{i_d}(x,m_1)u_1)\triangleright_{a_0} x_{i_j} \quad (3.1)$$

and we once again may argue by induction on degree. □

We shall use Proposition 3.4 to give a lower bound on the "size" of the free multi-Novikov algebra. To make sense of it, we define a $\mathbb{Z}\times\mathbb{N}$-grading on $\mathsf{CD}_A\langle X\rangle$ by postulating that each variable (x,m) has the grading

$$\left(-1+\sum_{a\in A} m(a), 1\right),$$

and extending this grading to products of variables additively. If we assume the sets A and X finite, the bi-homogeneous components of $\mathsf{CD}_A\langle X\rangle$ are finite-dimensional,[3] and Proposition 3.4 immediately implies the following lower bound.

Corollary 3.5. *For each $n \in \mathbb{N}$ the dimension of the degree n component of $\mathsf{Nov}_A\langle X\rangle$ is greater than or equal to the dimension of the $(-1,n)$-homogeneous component of $\mathsf{CD}_A\langle X\rangle$.*

We shall now obtain the same upper bound for the dimension of the degree n component of $\mathsf{Nov}_A\langle X\rangle$, which we shall use to prove Theorem 3.3. We shall for the moment perform all calculations in the multi-magmatic algebra $\mathsf{Mag}_A\langle X\rangle$ which has bilinear operations $\triangleright_a$, $a \in A$, without any identities between them. This algebra has an obvious basis of monomials which are binary trees whose internal vertices are labelled by elements of A and whose leaves are labelled by elements of X.

[3] We shall be making this assumption from now on; for the interested reader, we indicate that in the case where (say) X is infinite, one may consider a finer grading by $\mathbb{N}^X$ instead of $\mathbb{N}$, which allows for all our proofs to proceed without any problem.

The algebra $\mathsf{Mag}_A\langle X\rangle$ has an ideal I generated by all elements

$$(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) - (y \triangleright_a x) \triangleright_b z + y \triangleright_a (x \triangleright_b z), \tag{3.2}$$

$$(x \triangleright_a y) \triangleright_b z - x \triangleright_a (y \triangleright_b z) - (x \triangleright_b y) \triangleright_a z + x \triangleright_b (y \triangleright_a z), \tag{3.3}$$

$$(x \triangleright_a y) \triangleright_b z - (x \triangleright_b z) \triangleright_a y, \tag{3.4}$$

for all $a, b \in A$ and all $x, y, z \in \mathsf{Mag}_A\langle X\rangle$. The quotient by this ideal is clearly the free multi-Novikov algebra $\mathsf{Nov}_A\langle X\rangle$, and we shall now obtain a set of elements that spans the quotient $\mathsf{Mag}_A\langle X\rangle/I$.

Let $y_1, \ldots, y_{n+1} \in \mathsf{Mag}_A\langle X\rangle$ be some monomials. Recall that monomials $y_1 \triangleright_{a_1} (y_2 \triangleright_{a_2} (\ldots (y_n \triangleright_{a_n} y_{n+1}) \ldots))$ are often referred to as right-normed products of $y_1, \ldots, y_{n+1}$. Each monomial in $\mathsf{Mag}_A\langle X\rangle/I$ can be uniquely written as a right-normed product $y_1 \triangleright_{a_1} (y_2 \triangleright_{a_2} (\ldots (y_n \triangleright_{a_n} z) \ldots))$, where $z \in X$; for typographic purposes, we shall denote such product as $r(y_1, a_1, \ldots, y_n, a_n; z)$.

Definition 3.6. A right-normed product $r(y_1, a_1, \ldots, y_n, a_n; z)$ with $z \in X$ is said to be *left-leaning* if $y_2, \ldots, y_n \in X$, and y_1 is either a generator or a monomial whose right-normed product decomposition is itself left-leaning.

Let us furthermore specify a subclass of left-leaning products.

Definition 3.7. Let us fix an order of the set A and an order of the set X. We shall say that a left-leaning product $r(y_1, a_1, \ldots, y_n, a_n; z)$ is *ordered* if $n = 0$ (so that the whole product is just the generator z) or $n > 0$, $a_1 \le a_2 \le \cdots \le a_n$, and $y_1 \in X$ or $y_1 = r(y'_1, a'_1, \ldots, y'_m, a'_m; z')$ is an ordered left-leaning product

satisfying one of the two conditions:

- $n < m$,
- $n = m$ and $z < z'$.

Proposition 3.8. *The quotient* $\mathsf{Mag}_A\langle X\rangle/I$ *is spanned by cosets of ordered left-leaning products.*

Proof. Let us prove that in $\mathsf{Mag}_A\langle X\rangle/I$ every coset of a monomial u of degree d can be written as a linear combination of cosets of monomials of degree d which are ordered left-leaning products. We shall argue by induction on d, and for any fixed d by induction on n, the length of the right-normed product decomposition $r(y_1, a_1, \ldots, y_n, a_n; z)$ of our monomial u.

Clearly, Identity (2.7) implies that we can put $a_1, \ldots, a_n$ in the right order at the cost of elements with same d and smaller n that appear (to which the induction hypothesis applies). Similarly, Identity (2.6) implies that we can put $y_1, \ldots, y_n$ in the decreasing order according to their degrees at the cost of elements with same d and smaller n that appear (to which the induction hypothesis applies). If the resulting element

$$u' = r\left(z_1, b_1, \ldots, z_n, b_n; z\right)$$

satisfies $z_1 \in X$, then the fact that the degrees of $z_1, \ldots, z_n$ are in the decreasing order implies that we have $z_i \in X$ for all i, and so u' is an ordered right-leaning product. If $z_1 \notin X$, then the induction hypothesis applies to it, and its coset contains a linear combination of ordered right-leaning products. Without loss of generality, we may take

$$z_1 = r\left(z_1', b_1', \ldots, z_m', b_m'; z'\right)$$

with $z_2', \ldots, z_m' \in X$. Using Identity (2.8) for $a = b_1$ and $b = b_1'$, we see that in the quotient $\mathsf{Mag}_A\langle X\rangle/I$ we have

$$u' = r\left(r\left(z_1', b_1, z_2, \ldots, z_n, b_n; z\right), b_1', z_2', \ldots, z_m', b_m'; z'\right)$$

Now we can use the induction hypothesis to write the monomial

$$r(z_1', b_1, z_2, \ldots, z_n, b_n; z)$$

as a linear combination of ordered right-leaning products. To ensure the same for the monomial u', we may have to use Identity (2.8) once again, completing the proof. □

We are now ready to describe what will be a basis of the free multi-Novikov algebra. For that, we shall first recall and use some standard combinatorial argument (Knuth's rotation correspondence [Knu97, Chapter 2.3]) relating planar rooted binary trees and planar rooted trees. Let T be a planar rooted binary tree with n leaves. We define a planar rooted tree $\rho(T)$ with n vertices as follows: for each internal vertex i of T, consider the path going from i to the rightmost leaf of the tree growing from i, and collapse that path to a single vertex. An example of this is displayed in Figure 3.1, where we use the same label for all the vertices of the planar binary tree collapsed to the same one by the map ρ.

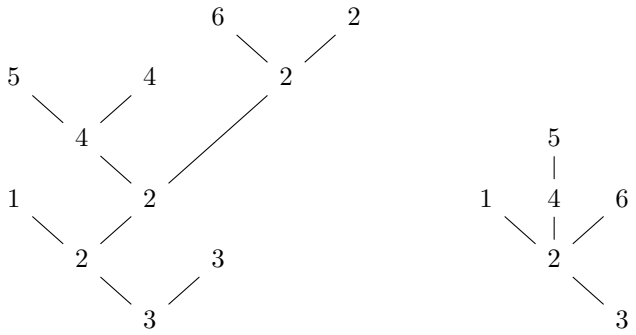


Figure 3.1: Knuth's rotation correspondence

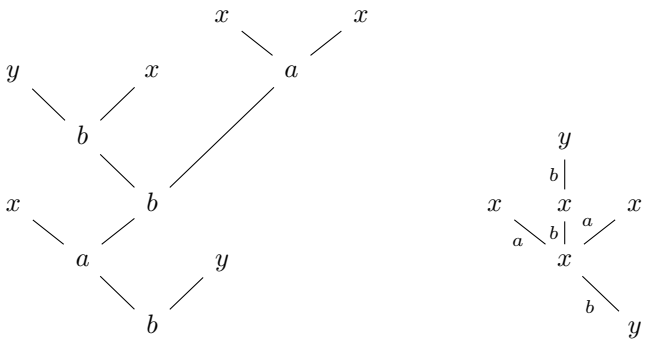


Figure 3.2: Knuth's rotation correspondence with labels

If T is the underlying planar binary tree of a monomial $u \in \mathsf{Mag}_A\langle X\rangle$, we have a labelling of all internal vertices of T by elements of the set A and a labelling of all leaves of T by elements of the set X. These labellings induce the labelling of vertices and edges of the planar tree $\rho(T)$: each vertex of $\rho(T)$ acquires the label of the leaf of T which we collapsed onto that vertex, and each edge of $\rho(T)$ acquires the label of the vertex of T for which that edge is the left incoming edge. An example of this is displayed in Figure 3.2.

We shall refer to an occurrence of a generator $x \in X$ in u as a *root* if it becomes a label of an internal vertex of $\rho(T)$, and a *leaf* if it becomes a label of a leaf of $\rho(T)$.

Definition 3.9. We shall call an ordered left-leaning product *normal* if the sequence of its leaves, read from the left to the right, is in the increasing order with respect to the order of X.

We are now ready to prove the main technical result of this section.

Proposition 3.10. *The quotient* $\mathsf{Mag}_A\langle X\rangle/I$ *is spanned by cosets of normal ordered left-leaning products.*

Proof. Thanks to the result of Proposition 3.8, it is enough to prove that every ordered left-leaning product can be written as a linear combination of cosets of normal ordered left-leaning products. Let us take an ordered left-leaning product

$$u = r(y_1, a_1, \ldots, y_n, a_n; z)$$

with $a_1 \leq a_2 \leq \cdots \leq a_n$, $y_2, \ldots, y_n \in X$, and either $y_1 \in X$ or

$$y_1 = r\left(y'_1, a'_1, \ldots, y'_m, a'_m; z'\right)$$

satisfies $n < m$ or $n = m$ and $z < z'$. We shall prove that in $\mathsf{Mag}_A\langle X\rangle/I$ it can be written as a linear combination of cosets of normal ordered left-leaning products by induction on d, and for a fixed d by induction on ℓ, the number of leaves, and for fixed d and ℓ by the sum $n + m$.

Suppose first that $y_1 \in X$, so that all the elements $y_1, \ldots, y_n$ are leaves. Using Identity (2.6), we may put them in order at the cost of elements with fewer leaves that appear. We may apply the procedure of Proposition 3.8 (which does not increase the number of leaves) to these elements, and then the induction hypothesis finishes the argument.

Suppose now that $y_1 \notin X$, so that we have $y_1 = r(y'_1, a'_1, \ldots, y'_m, a'_m; z')$ with $n < m$ or $n = m$ and $z < z'$. By the induction hypothesis we may assume that y_1 is a normal ordered left-leaning product. Also, modulo terms with fewer leaves, we may assume $y_2 \leq y_3 \leq \cdots \leq y_n$. Suppose $y'_m > y_2$. If we show that these two elements can be exchanged modulo I, the proof will be finished after doing several exchanges like that.

Using our relations, we can change the order from $r(y'_1, a'_1, \ldots, y'_m, a'_m; z')$ to $r(y'_m, a'_1, y'_1 \ldots, y'_{m-1}, a'_m; z')$ at the cost of some elements with fewer leaves, and one more element $r((y'_1 \triangleright a'_1 y'_m), a_2, \ldots, y'_{m-1}, a'_m; z')$ with the same number of leaves. However, this element has fewer factors in the second level right-normed factorisation, and the induction on $n + m$ takes care of that. Thus, we may focus on the element

$$r\left(r(y'_m, a'_1, y'_1 \ldots, y'_{m-1}, a'_m; z'), a_1, y_2, \ldots, y_n, a_n; z\right),$$

which, thanks to Identity (2.8), is equal to

$$r\left(r(y'_m, a_1, y_2, \ldots, y_n, a_n; z), a'_1, y'_1 \ldots, y'_{m-1}, a'_m; z'\right),$$

Now, as before, we use Identity (2.6) to exchange in $r(y'_m, a_1, y_2, \dots, y_n, a_n; z)$ the elements y'_m and y_2, and use Identity (2.8) again to obtain

$$r\left(r(y_2, a'_1, y'_1 \dots, y'_{m-1}, a'_m; z'), a_1, y'_m, \dots, y_n, a_n; z\right),$$

and an argument similar to before allows us to exchange y_2 and y'_1 in the inner monomial $r(y_2, a'_1, y'_1, \dots, y'_{m-1}, a'_m; z')$, leading to a linear combination of normal ordered left-leaning products by the induction hypothesis. □

Proof of Theorem 3.3. Let us show that normal ordered left-leaning products of degree d are in bijection with the standard basis elements of the $(-1, d)$-homogeneous component of $\mathsf{CD}_A\langle X\rangle$. The way to construct a basis element out of an ordered left-leaning product is quite straightforward: to each root vertex labelled by x whose operations in the right-normed factorisation are $\triangleright_{a_1}, \dots, \triangleright_{a_n}$ with $a_1 \leq a_2 \leq a_n$, we associate the variable $\partial_{a_1} \dots \partial_{a_n}(x)$, and to each leaf labelled by x we associate simply the variable x. The reverse reconstruction is straightforward (the ordered condition of the left-leaning product ensures uniqueness), and leads to a valid monomial in $\mathsf{Mag}_A\langle X\rangle$ thanks to the populated condition.

Since dimensions of bi-homogeneous components of $\mathsf{CD}_A\langle X\rangle$ are finite-dimensional, we see that the surjective map

$$\mathsf{Nov}_A\langle X\rangle \twoheadrightarrow \mathsf{CD}_A\langle X\rangle_{-1}$$

must be an isomorphism, as required. □

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